

MATHWIZURD (/)

Apr 20

Integrals

with

Functions

as Bounds

Fundamental Theorem of Calculus

There are two parts of the Fundamental Theorem of
Calculus:

Part One 定积分

$$\int_a^b f(x) dx = F(a) - F(b) \text{ where } F(x) \text{ is the antiderivative of } f(x)$$

Part Two 不定积分

$$\text{If } F(x) = \int_a^x f(t) dt, \text{ then } \frac{d}{dx} F(x) = f(x)$$

Integral with Functions as

Bounds

ONE BOUND

一个可变上限

上界为函数, 则代入该函数后再对该函数求导

$$\text{Calculate } \frac{d}{dx} \int_a^{h(x)} f(t) dt$$

$$\text{Let } G(x) = \int_a^x f(t) dt$$

$$G(h(x)) = \int_a^{h(x)} f(t) dt$$

$$\begin{aligned} \frac{d}{dx} G(h(x)) &= G'(h(x)) h'(x) \\ &= f(h(x)) h'(x) \end{aligned}$$

TWO BOUNDS

上下均为变量

$$\text{Calculate } \frac{d}{dx} \int_{g(x)}^{h(x)} f(t) dt$$

$$\int_a^{h(x)} f(t) dt + \int_{g(x)}^a f(t) dt$$

$$\text{Just split it up: } \frac{d}{dx} \left(- \int_a^{g(x)} f(t) dt + \int_a^{h(x)} f(t) dt \right)$$

$$\begin{aligned} &-f(g(x))g'(x) + f(h(x))h'(x) \\ &= f(h(x))h'(x) - f(g(x))g'(x) \end{aligned}$$

NO BOUNDS

The derivative is 0, because that's just a constant.

Examples

$$\text{Calculate } \frac{d}{dx} \int_x^{x^3} \sin(t) dt$$

Let F be the antiderivative of $\sin(t)$. So, this expression equals

$$\begin{aligned} &\frac{d}{dx} (F(x^3) - F(x)) \\ &3x^2 F'(x^3) - F'(x) \end{aligned}$$

Recall F is the antiderivative of $\sin(t)$, so $F' = \sin(t)$.

$$\boxed{3x^2 \sin(x^3) - \sin(x)}$$

$$\text{Calculate } \frac{d}{dx} \int_{\sqrt{x}}^{x^2} \sqrt{t} \cos(t) dt$$

Let F be the antiderivative of $\sqrt{t} \cos(t)$. This expression equals:

$$\begin{aligned} &\frac{d}{dx} (F(x^2) - F(\sqrt{x})) \\ &2x F'(x^2) - \frac{1}{2\sqrt{x}} * F'(\sqrt{x}) \end{aligned}$$

$$2x * \sqrt{x^3} \cos(x^2) - \frac{1}{2\sqrt{x}} x^{1/4} \cos(\sqrt{x})$$

$$2x^2 \cos(x^2) - \frac{1}{2} x^{-1/4} \cos(\sqrt{x})$$

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Apr 11 Testing for Convergence and Divergence

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